SUSHIRMALA MAHILA BACHAT GAT

Aditya Patil, Ninad Patil, Piyush Patil

Department of Information Technology Vidyavardhini's College of Engineering and Technology Vasai. India

Email : aditya.233354105@vcet.edu.in
ninad.233374105@vcet.edu.in
piyush.233384101@vcet.edu.in

I. Abstract:

" This mini-project presents the design and development of "Sushrimala Mahila Bachat Gat," a comprehensive web application built using the MERN stack.The platform digitizes the operations of women's self-help groups (SHGs), focusing on community savings, loan management, and financial empowerment. It provides a secure, role-based system for managing member contributions, loan applications, group meetings, and membership enquiries. By automating core administrative tasks, the application enhances transparency, reduces manual effort, and promotes financial inclusion among women in rural and urban communities.

II. Introduction :

In many communities, self-help groups (SHGs) play a pivotal role in promoting savings, providing credit, and fostering collective decision-making among women. However, managing these groups often involves maintaining physical records, which can be prone to errors, lack transparency, and are difficult to scale.

This project aims to address these challenges by developing a digital platform that automates and streamlines the core operations of a Bachat Gat (savings group). The system enables secure user management, tracks financial contributions, processes loan applications, and facilitates communication between members and group officers.

III. Problem Statement:

Traditional record-keeping for self-help groups is largely manual, leading to inefficiencies, potential discrepancies in financial records, and limited accessibility for members. There is a need for a centralized, transparent, and easy-to-use digital system that can manage member data, contributions, loans, and group activities efficiently, thereby empowering women through technology.

IV. Description :

A. Key Concepts

The development of the Mahila Bachat Gat website involves several core concepts:

1.Role-Based Access Control (RBAC): The system defines distinct user roles (Member, President, Secretary, Treasurer, Admin) with specific permissions to ensure data security and operational integrity.

2.RESTful API Architecture: A structured backend provides endpoints for all frontend operations, enabling secure data exchange between the client and server.

3.State Management: The frontend, built with React, efficiently manages the application state for a dynamic and responsive user experience.

4.Financial Transaction Integrity: The system ensures accuracy in recording and tracking all financial activities, including contributions and loan repayments.Key features of the system include:

B. System Architecture

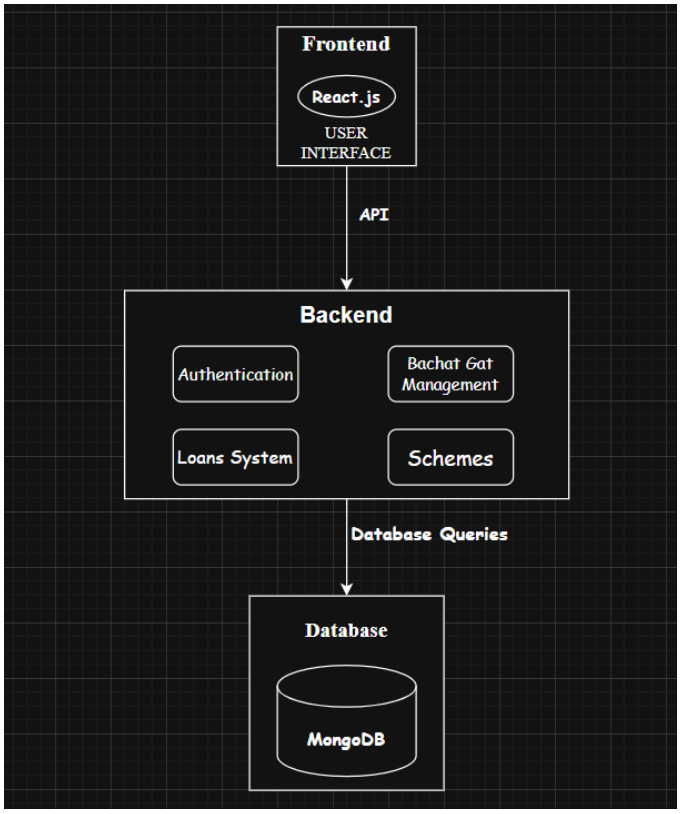
The application follows a client-server architecture with the following components:

1.Frontend (Client): Built with React and TypeScript, this module provides the user interface for all interactions. It handles user input, displays data, and communicates with the backend via HTTP requests.

2.Backend (Server): Developed using Node.js and Express.js, this module processes business logic, handles authentication, and interacts with the database. It exposes a REST API for the frontend.

3.Database: MongoDB is used as the primary database to store all structured and semi-structured data, including user profiles, group details, financial records, and loan applications.

4.Authentication Layer: JWT (JSON Web Tokens) is implemented to secure API endpoints and manage user sessions**.**



V. Methodology:

The system operates through a structured workflow:

1.User Interaction: A user accesses the web application via a browser.

2.Authentication & Authorization: The user logs in, and the system verifies their credentials and role.

3.Request Handling: The frontend sends an HTTP request to the appropriate backend API endpoint based on the user's action.

4.Business Logic Execution: The backend server processes the request, which may involve querying or updating the database, applying business rules (e.g., loan approval checks).

5.Data Persistence: All changes are saved in the MongoDB database.

6.Response Generation: The backend sends a structured response (e.g., JSON data, success message) back to the frontend.

7.UI Update: The frontend receives the response and updates the user interface accordingly, displaying the requested information or confirming the action.

VI. Result:

The developed platform successfully meets its objectives. The following features have been implemented and tested:

-A secure user registration and login system with JWT-based authentication.

-A dashboard for members to view their profile, contribution history, and group details.

-An administrative interface for group officers (President, Secretary, Treasurer) to manage members, record contributions, and process loan applications.

-A public enquiry form for new members to express interest in joining a group.

-A responsive and multilingual user interface (English, Hindi, Marathi).

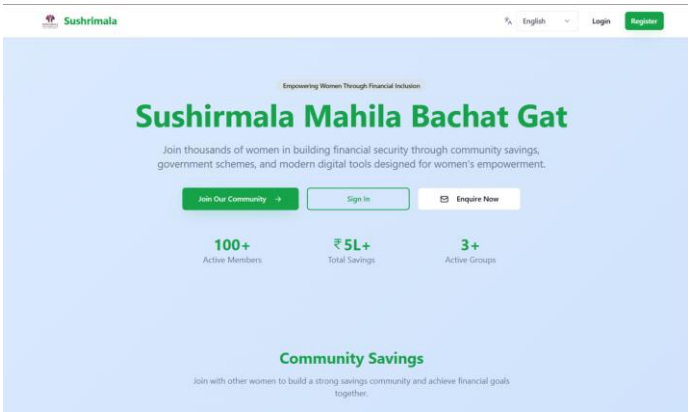


Fig1.Home_Page

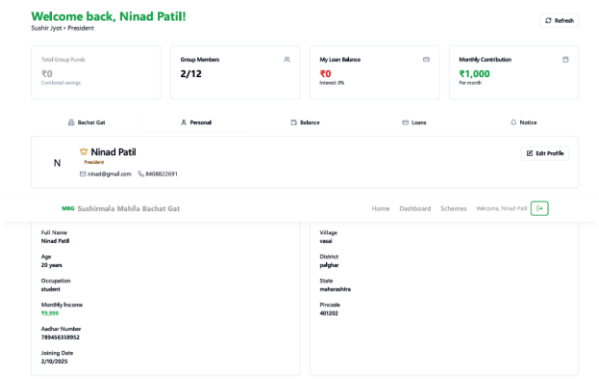


Fig3.Personal_Info_Page

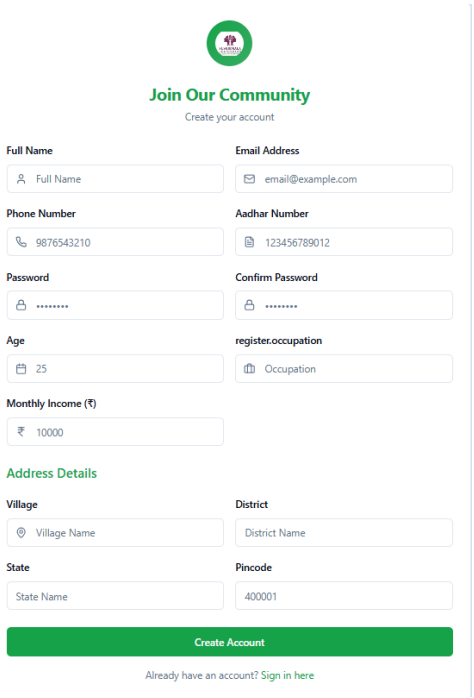


Fig2.Register/Login Page

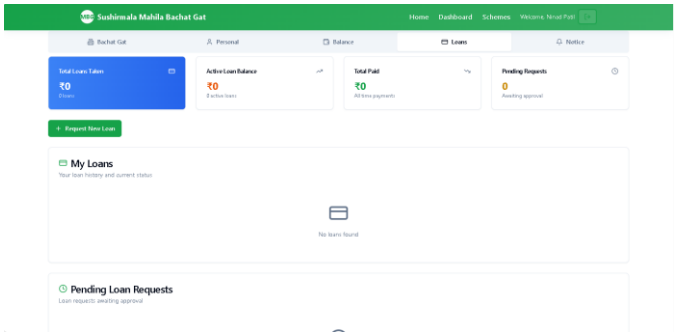


Fig4.Loan_Page

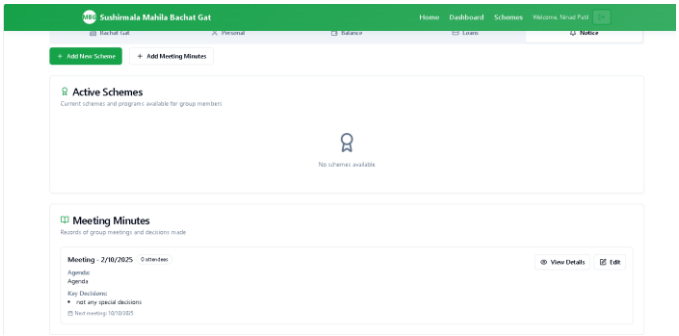
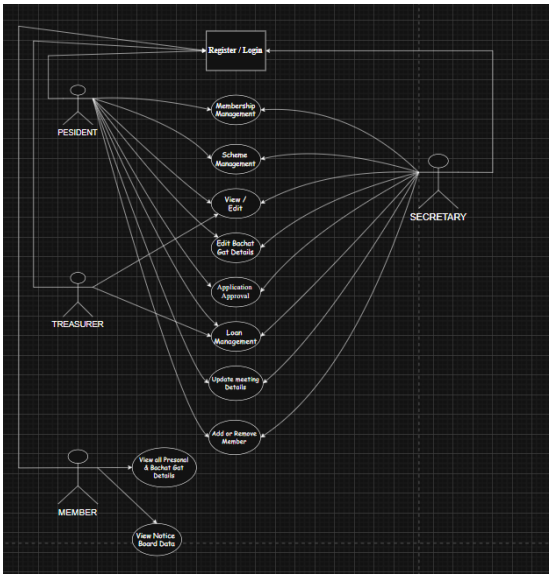


Fig5.Notice_Page

VII. Use Case Diagram:



VIII. Conclusion:

The "Sushrimala Mahila Bachat Gat" website demonstrates the effective application of the MERN stack to solve a real-world problem. The platform offers a robust set of features:

-User and Group Management: Efficient handling of members and group hierarchies.

-Financial Tracking: Accurate recording and visualization of member contributions and group funds.

-Loan Management: A streamlined workflow for loan applications, approvals, and tracking.

-Enquiry System: A digital channel for membership growth.

-Role-Based Access: A secure system ensuring users access only relevant information and functions.

-This project lays a foundation for future enhancements, such as integrating payment gateways for online contributions, adding data analytics dashboards, and developing a mobile application for greater accessibility.

IX. Future Scope:

The current platform provides a strong foundation for managing self-help group operations. To further enhance its functionality, accessibility, and impact, the following features are planned for future implementation:

1. Integration of Payment Gateway

A primary future enhancement is the integration of a secure payment gateway (such as Razorpay, Stripe, or Paytm). This will allow members to make their monthly contributions and loan repayments directly through the platform via UPI, net banking, debit/credit cards, or digital wallets. Key benefits include:

Convenience: Members can contribute from anywhere, reducing reliance on cash and physical meetings.

Automated Record-Keeping: Successful transactions will be automatically logged in the system, updating the member's contribution status and the group's total funds in real-time, thereby minimizing manual entry errors.

Financial Transparency: A digital trail for every transaction will provide an immutable record, enhancing trust among group members.

2. Mobile Application Development

Developing a dedicated cross-platform mobile application (using frameworks like React Native or Flutter) will significantly improve accessibility. A mobile app would offer:

Push Notifications: For reminders about upcoming meetings, contribution deadlines, loan payment dues, and important announcements.

Offline Functionality: Basic features like viewing profile details and contribution history could be made available offline.

Broader Reach: Easier access for users in areas with limited desktop or laptop usage.